

Direct and inverse variation



1 $k = 2$

2 a $M = kN^2$

b $k = 4$

3 a $y = kx^3$

b $k = 3$

4 a $y = kx^n$

b $k = 8$

5 a $y = kx^n$

b $k = -6$

6 $k = 2$

7 $k = \frac{3}{4}$

8 a $m = \frac{1}{4}$

b $k = 2$

Inverse variation

9 $y = \frac{k}{x}$

10 $k = 150$

11 a Direct

b Inverse

12 $r = \frac{k}{a}$

13 a No

b No

c No

d Yes

e Yes

f No

14 a 375

b $\frac{125}{2}$



c $\frac{125}{4}$

15 a Yes

b Yes

c No

d No

16 a $k = 252$

b $x = 9, y = 42$

17 a $t = \frac{48}{s}$

b $t = \frac{2}{s^2}$

18 a 0

b 0

Applications

19 a 1.27 AUD/USD

b 0.79 USD/AUD

20 a $H = kM$

b $k = 0.006$

c $H = 0.342$

d $M = 14$

21 a $C = k\sqrt[4]{B}$

b $k = 17.4$

c $C = 43.8$

d $B = 2.6$